Number of contacts: comparing the m-modes

In the paper every active agent contacts the same number of agents, $m = 10$. Our NetLogo model (`netlogo/polarisation.nlogox`) adds a setting, **m-mode**, that lets this number depend on how extreme an agent's opinion is. The mean over all agents always stays m; the modes only change who gets how many contacts:

m-mode	who gets more contacts
uniform	nobody: everyone contacts m agents, as in the paper
boost-extreme	extreme agents, up to `m-multiplier` $= 6.7$ times as many as neutral ones
uprank-bridging and downrank-toxic	neutral agents, up to 6.7 times as many as extreme ones
uprank-bridging	like the previous one, but the 20% most extreme agents keep the number they had before they became that extreme

The Info tab of the model explains the modes in detail.

This notebook compares the four modes in the three regimes of Fig. 1 of the paper, using two measures taken at the end of each run:

- $|\langle x \rangle|$ (`abs-mean-opinion`), the absolute mean opinion. It is large when everyone ends up on one side, but close to 0 both at consensus and when the population is split into two equal halves.
- $\mathrm{Var}(x)$ (`opinion-variance`), the variance of the opinions. It is close to 0 only at consensus.

Input and output:

- input: `data/m-modes/m-modes.csv`, the spreadsheet output of the BehaviorSpace experiment `m-modes`
- output: `results/m-modes/`, a summary table and the plots

```
input_file <- "../data/m-modes/m-modes.csv"
output_folder <- "../results/m-modes"
dir.create(output_folder, recursive = TRUE, showWarnings = FALSE)
```

The data

The spreadsheet output of BehaviorSpace starts with 6 lines about the experiment. After that, every row is one parameter or measure, and every run has four columns next to each other: the step at which the run ended and the three measures, in the order they were entered in BehaviorSpace. The first column holds the row names, such as `alpha` or `m-mode`.

```
sheet <- read.csv(input_file, skip = 6, header = FALSE, colClasses = "character")
row_names <- sheet[, 1]
sheet <- sheet[, -1]
```

The row `[final value]` names what is in each column (`[step]` or one of the measures), and the row below it holds the values at the end of the run. The parameters are written only in a run's first column, the one named `[step]`; the other three are empty. Two small functions pick out a parameter or a measure for all runs:

```
final_names <- unlist(sheet[row_names == "[final value]", ])
final_values <- unlist(sheet[which(row_names == "[final value]") + 1, ])
first_column <- final_names == "[step]"

parameter <- function(name) unname(unlist(sheet[row_names == name, first_column]))
final_value <- function(name) as.numeric(final_values[final_names == name])

runs <- data.frame(
  alpha = as.numeric(parameter("alpha")),
  beta = as.numeric(parameter("beta")),
  mode = parameter("m-mode"),
  t_end = final_value("[step]") * as.numeric(parameter("dt")),
  abs_mean = final_value("abs-mean-opinion"),
  variance = final_value("opinion-variance"),
  mean_contacts = final_value("mean [ n-contacts ] of turtles")
)
head(runs)
```

```
##   alpha beta    mode t_end     abs_mean     variance mean_contacts
## 1  0.05    0 uniform    10 2.400275e-06 6.891071e-10            10
## 2  0.05    0 uniform    10 1.200634e-06 6.847047e-10            10
## 3  0.05    0 uniform    10 6.609423e-07 6.897835e-10            10
## 4  0.05    0 uniform    10 7.611842e-07 6.906545e-10            10
## 5  0.05    0 uniform    10 2.761479e-06 6.915350e-10            10
## 6  0.05    0 uniform    10 3.332265e-06 6.930529e-10            10
```

The mean number of contacts should be $m = 10$ in every run, whatever the mode:

```
table(runs$mean_contacts)
```

```
##
##   10
## 2400
```

All runs end at the same time:

```
t_end <- unique(runs$t_end)
t_end
```

```
## [1] 10
```

The experiment also contains combinations of α and β that we don't use. We keep only the three regimes of Fig. 1 (`merge()` keeps the runs whose α and β appear in `regimes`):

```r
regimes <- data.frame(
  alpha = c(0.05, 3, 3),
  beta = c(2, 0, 3),
  regime = c("consensus", "one-sided radicalization", "polarization")
)
runs <- merge(runs, regimes)

modes <- c("uniform", "boost-extreme", "uprank-bridging and downrank-toxic", "uprank-bridging")
runs$mode <- factor(runs$mode, levels = modes)
runs$regime <- factor(runs$regime, levels = regimes$regime)

run_counts <- table(runs$regime, runs$mode)
run_counts
```

```
##
##                            uniform boost-extreme
##   consensus                    100           100
##   one-sided radicalization     100           100
##   polarization                 100           100
##
##                            uprank-bridging and downrank-toxic uprank-bridging
##   consensus                                               100             100
##   one-sided radicalization                                100             100
##   polarization                                            100             100
```

Every plot below states when the runs ended and how many runs there are per mode:

```r
runs_per_mode <- paste(unique(as.vector(run_counts)), collapse = " or ")
run_info <- paste0("t = ", t_end, ", ", runs_per_mode, " runs per m-mode")
run_info
```

```
## [1] "t = 10, 100 runs per m-mode"
```

Summary table

For each regime and mode: the mean and standard deviation over the runs, and how many times larger the mean is than under uniform (the paper's model). A value of 2 means twice as large as uniform, 0.5 half as large.

```r
means <- aggregate(cbind(abs_mean, variance) ~ regime + mode, data = runs, FUN = mean)
sds <- aggregate(cbind(abs_mean, variance) ~ regime + mode, data = runs, FUN = sd)

summary_table <- data.frame(
  regime = means$regime,
  mode = means$mode,
  abs_mean = means$abs_mean,
  abs_mean_sd = sds$abs_mean,
  variance = means$variance,
  variance_sd = sds$variance
)
```

```
# divide by the uniform mean of the same regime
uniform <- summary_table[summary_table$mode == "uniform", ]
row_of_uniform <- match(summary_table$regime, uniform$regime)
summary_table$abs_mean_vs_uniform <- summary_table$abs_mean / uniform$abs_mean[row_of_uniform]
summary_table$variance_vs_uniform <- summary_table$variance / uniform$variance[row_of_uniform]

summary_table <- summary_table[order(summary_table$regime, summary_table$mode), ]
write.csv(summary_table, file.path(output_folder, "summary.csv"), row.names = FALSE)

numbers <- sapply(summary_table, is.numeric)
summary_table[numbers] <- signif(summary_table[numbers], 3)
knitr::kable(summary_table, row.names = FALSE)
```

regime	mode	abs_mean	abs_mean_sd	variance	variance_sd	abs_mean_vs_uniform	variance_vs_uniform
consensus	uniform	1.75e-05	1.57e-05	0.00e+00	0.00e+00	1.0000	1.000
consensus	boost-extreme	7.86e-04	1.32e-03	6.80e-06	4.78e-05	44.8000	239.000
consensus	uprank-bridging and downrank-toxic	1.60e-06	1.20e-06	0.00e+00	0.00e+00	0.0916	0.150
consensus	uprank-bridging	4.40e-06	3.40e-06	0.00e+00	0.00e+00	0.2490	0.240
one-sided radicalization	uniform	1.85e+00	1.16e-01	1.41e+00	3.33e-01	1.0000	1.000
one-sided radicalization	boost-extreme	2.54e+00	2.67e-01	1.36e+01	4.19e+00	1.3700	9.660
one-sided radicalization	uprank-bridging and downrank-toxic	1.40e+00	6.04e-02	2.31e-01	3.70e-02	0.7530	0.164
one-sided radicalization	uprank-bridging	1.76e+00	9.05e-02	1.06e+00	2.14e-01	0.9480	0.749
polarization	uniform	8.99e-02	7.53e-02	7.47e+01	1.34e+00	1.0000	1.000
polarization	boost-extreme	9.80e-01	4.39e-01	2.38e+01	5.56e+00	10.9000	3.190
polarization	uprank-bridging and downrank-toxic	2.59e-02	1.80e-02	2.45e+00	2.54e-01	0.2880	0.328
polarization	uprank-bridging	5.80e-02	4.27e-02	5.89e+00	9.30e-01	0.6450	0.788

Plots

All plots use the same colour for each mode, and put the mode names on two lines where they are long:

```
mode_colours <- c("grey40", "firebrick", "steelblue", "darkorange")
mode_labels <- c("uniform", "boost-extreme", "uprank-bridging\nand downrank-toxic", "uprank-bridging")

# plot title with the values of alpha and beta of a regime
regime_title <- function(regime) {
  r <- regimes[regimes$regime == regime, ]
  bquote(alpha == .(r$alpha) * "," ~~ beta == .(r$beta))
}
```

Each measure per mode

One figure per measure, with one panel per regime. Each coloured dot is one run. Next to them, the black dot is the mean and the bar the mean ± one standard deviation. The dashed line is the uniform mean, so a mode above the line increases the measure compared to the paper's model.

The values span several orders of magnitude (from about 10^{-10} at consensus to about 40), so the y-axis is logarithmic. A log axis can't show values ≤ 0, so where mean − sd is below the lowest run, the bar is cut off at the bottom of the plot.

```
plot_measure <- function(measure, label, file) {
  values <- runs[[measure]]
  png(file, width = 1500, height = 600, res = 110)
  par(mfrow = c(1, 3), mar = c(10.5, 4.5, 5, 1), oma = c(0, 0, 2.5, 0))

  for (regime in levels(runs$regime)) {
    in_regime <- runs$regime == regime
    mode_number <- as.integer(runs$mode[in_regime])

    plot(jitter(mode_number - 0.1, amount = 0.08), values[in_regime],
      log = "y", ylim = range(values),
      xlim = c(0.5, 4.5), xaxt = "n", xlab = "", ylab = label,
      pch = 19, cex = 0.6, col = mode_colours[mode_number]
    )
    axis(1, at = 1:4, labels = mode_labels, las = 2)
    title(regime_title(regime), line = 2.5, cex.main = 1.5)
    mtext(regime, side = 3, line = 0.5)

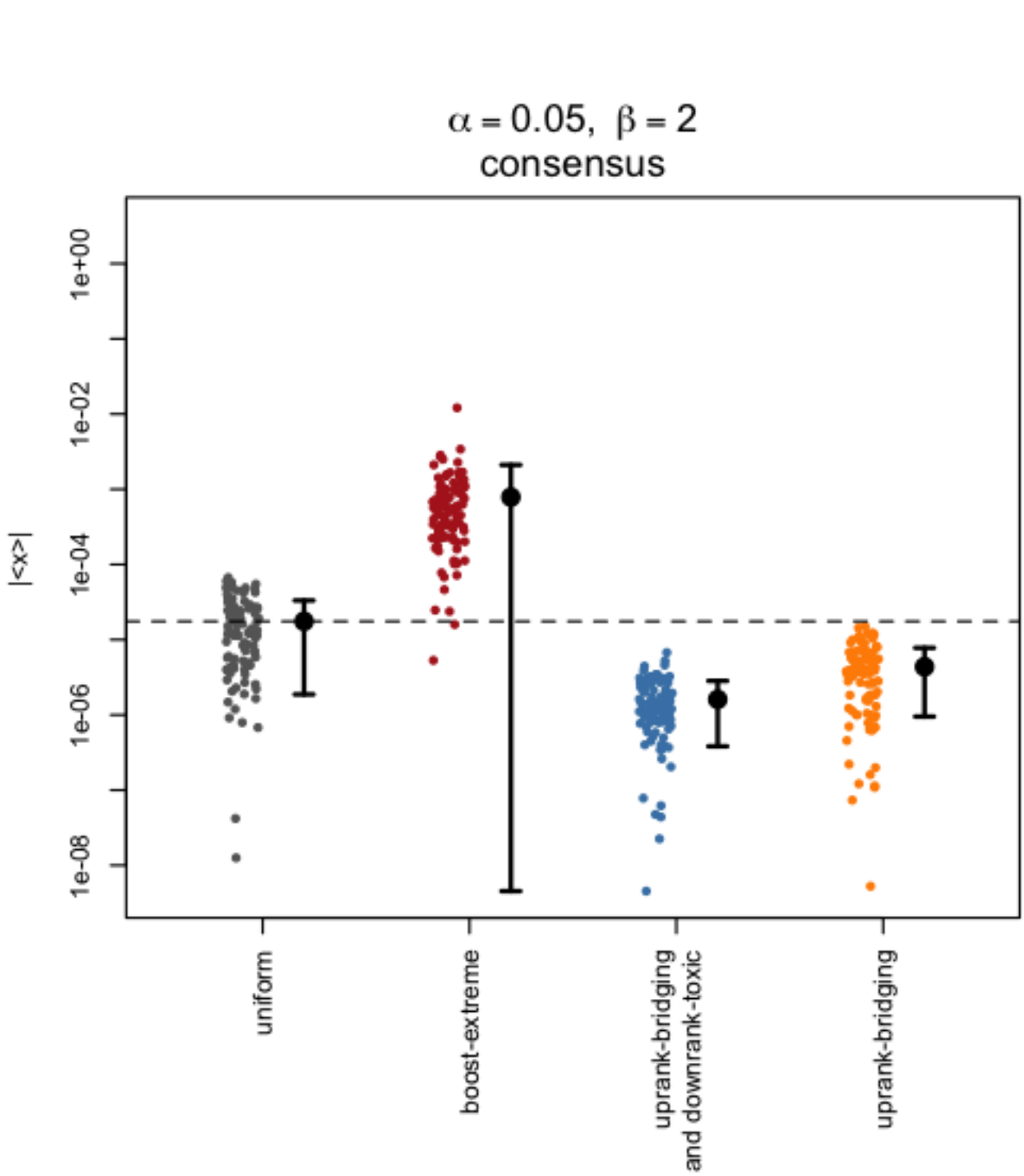
    mean_per_mode <- tapply(values[in_regime], runs$mode[in_regime], mean)
    sd_per_mode <- tapply(values[in_regime], runs$mode[in_regime], sd)
    bar_bottom <- pmax(mean_per_mode - sd_per_mode, min(values))
    arrows(1:4 + 0.2, bar_bottom, 1:4 + 0.2, mean_per_mode + sd_per_mode,
      angle = 90, code = 3, length = 0.04, lwd = 2
    )
    points(1:4 + 0.2, mean_per_mode, pch = 19, cex = 1.5)
    abline(h = mean_per_mode["uniform"], lty = 2)
  }

  mtext(paste0(label, " at the end of the run (", run_info, ")"),
    outer = TRUE, cex = 1.1
  )
  dev.off()
}
```

```
plot_measure("abs_mean", "|<x>|", file.path(output_folder, "abs-mean-opinion.png"))
```

```
## pdf
##   2
```

```
knitr::include_graphics(file.path(output_folder, "abs-mean-opinion.png"))
```

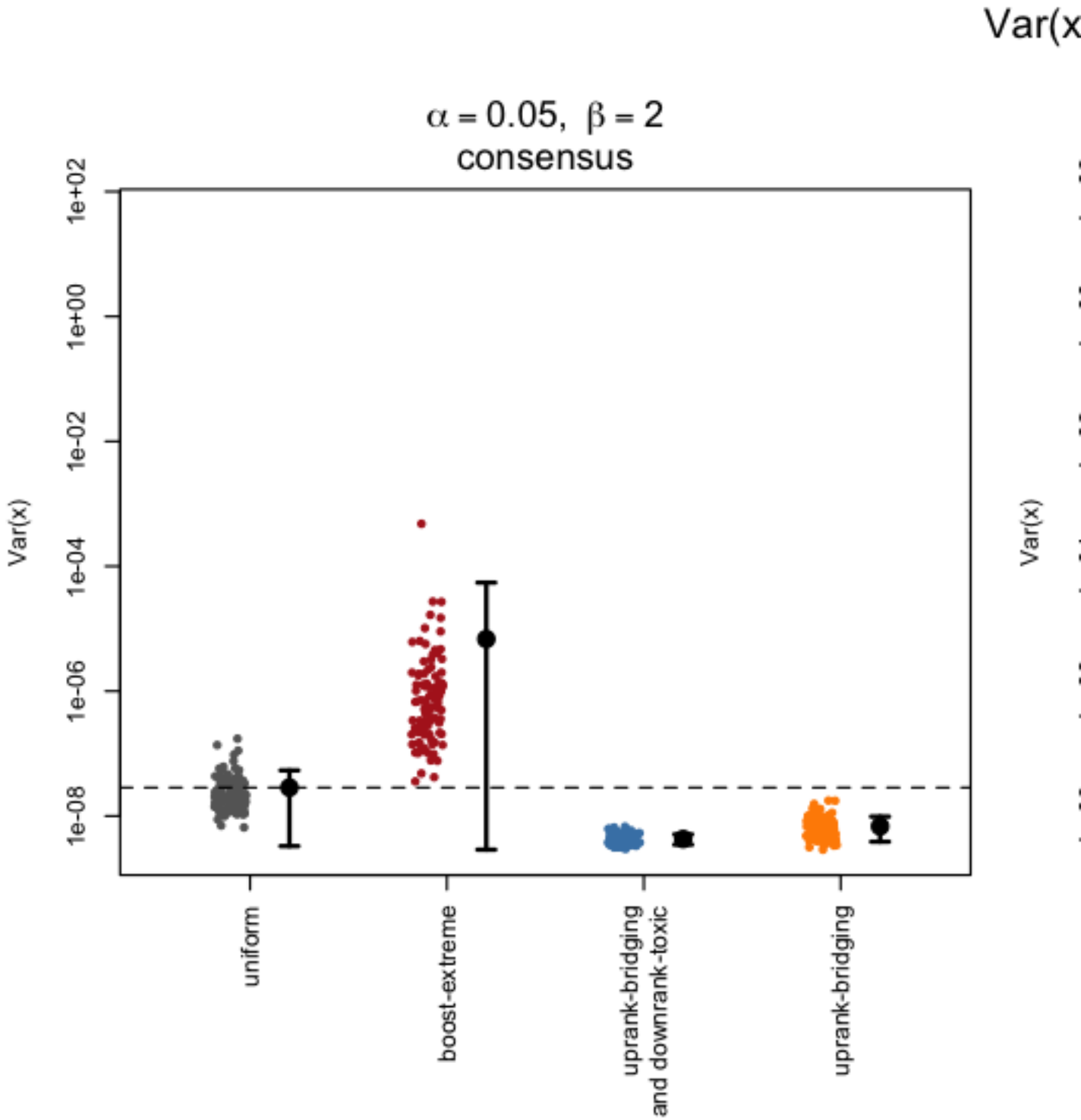



```
plot_measure("variance", "Var(x)", file.path(output_folder, "opinion-variance.png"))
```

```
## pdf
```

```
##   2
```

```
knitr::include_graphics(file.path(output_folder, "opinion-variance.png"))
```



Distribution of the variance

The dot plots above summarise each mode by its mean. A violin plot shows the whole distribution of the variance over the runs: the width of a violin shows how many runs ended with that variance, like a histogram turned on its side and smoothed. The box inside is a boxplot (thick line: median; box: middle 50% of the runs), and the dots are the individual runs.

There is one plot per regime, each with its own (logarithmic) y-axis, because the regimes differ too much to share one. The plots are wide (16:9) so they fit on a slide.

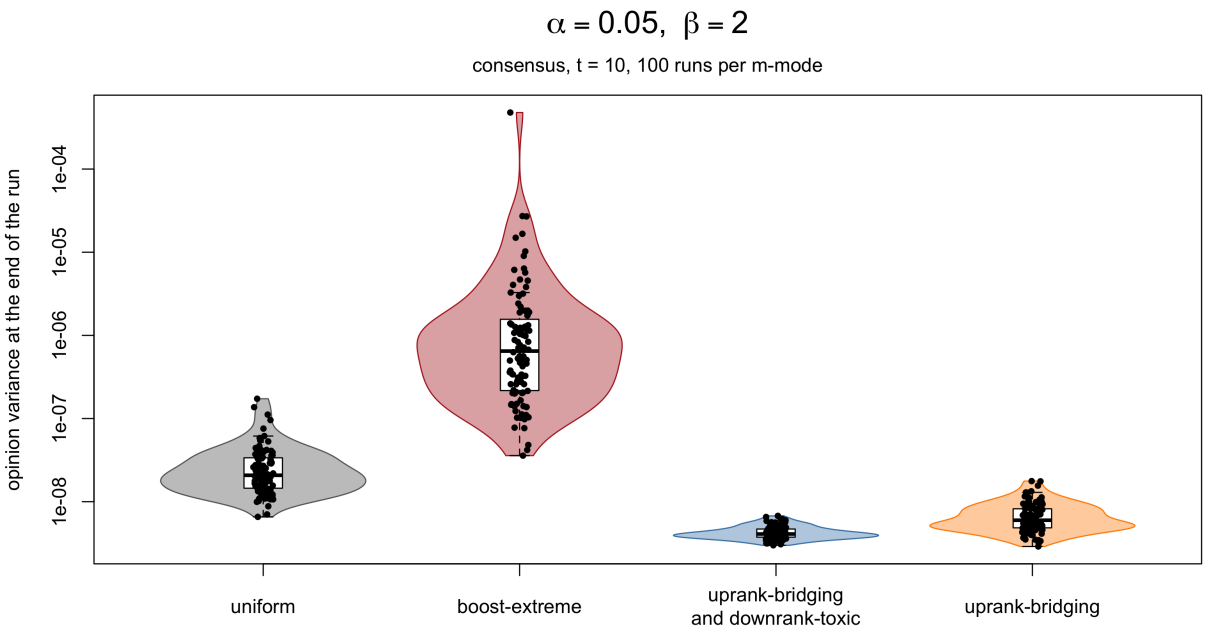
```
plot_violins <- function(regime, file) {
  in_regime <- runs$regime == regime
  variance <- runs$variance[in_regime]
  mode <- runs$mode[in_regime]

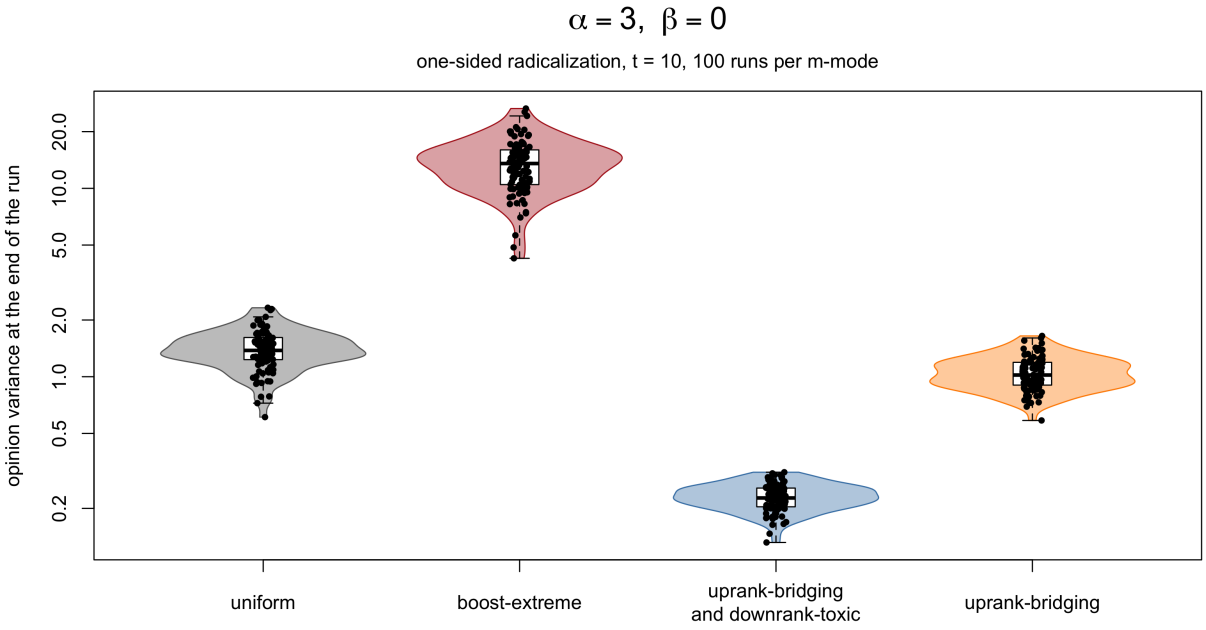
  png(file, width = 3200, height = 1800, res = 300)
  par(mar = c(5, 5, 5, 1))
  plot(NA,
    log = "y", xlim = c(0.5, 4.5), ylim = range(variance),
    xaxt = "n", xlab = "", ylab = "opinion variance at the end of the run"
  )
  axis(1, at = 1:4, labels = mode_labels, padj = 0.5)
  title(regime_title(regime), line = 3, cex.main = 1.6)
  mtext(paste0(regime, ", ", run_info), side = 3, line = 0.8)

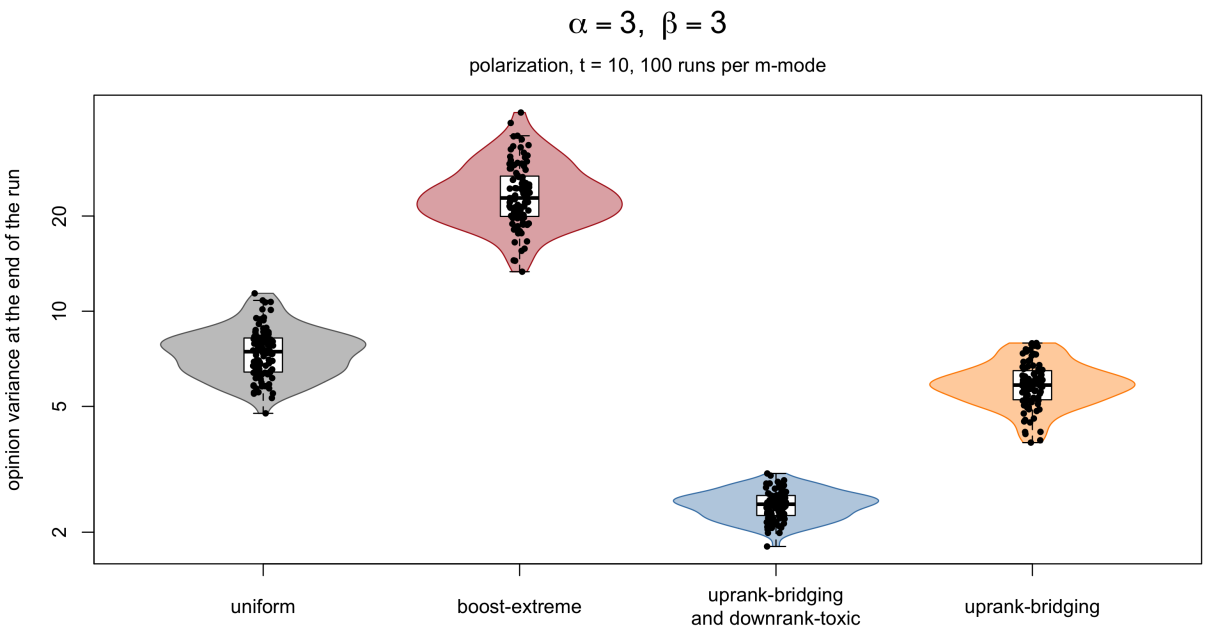
  # the violin is the density of log10(variance), drawn to both sides;
  # cut = 0 makes it end at the smallest and largest run
  for (k in 1:4) {
    d <- density(log10(variance[mode == modes[k]]), cut = 0)
    half_width <- 0.4 * d$y / max(d$y)
    polygon(c(k - half_width, rev(k + half_width)), 10^c(d$x, rev(d$x)),
      col = adjustcolor(mode_colours[k], alpha.f = 0.4), border = mode_colours[k]
    )
  }

  boxplot(variance ~ mode,
    at = 1:4, add = TRUE, boxwex = 0.15,
    outline = FALSE, axes = FALSE, col = "white"
  )
  points(jitter(as.integer(mode), amount = 0.04), variance, pch = 19, cex = 0.6)
  dev.off()
}
```

```
violin_files <- file.path(output_folder, paste0(
  "variance-violin-alpha", regimes$alpha, "-beta", regimes$beta, ".png"
))
for (k in 1:3) {
  plot_violins(regimes$regime[k], violin_files[k])
}
knitr::include_graphics(violin_files)
```


α = 0.05, β = 2
consensus, t = 10, 100 runs per m-mode
opinion variance at the end of the run
1e-08
1e-07
1e-06
1e-05
1e-04
uniform
boost-extreme
uprank-bridging and downrank-toxic
uprank-bridging


α = 3, β = 0
one-sided radicalization, t = 10, 100 runs per m-mode
opinion variance at the end of the run
0.2
0.5
1.0
2.0
5.0
10.0
20.0
uniform
boost-extreme
uprank-bridging and downrank-toxic
uprank-bridging


α = 3, β = 3
polarization, t = 10, 100 runs per m-mode
opinion variance at the end of the run
2
5
10
20
uniform
boost-extreme
uprank-bridging
and downrank-toxic
uprank-bridging